Final Year Project Proposal: IoT-Based Smart Water Quality Monitoring System for Detecting Mining Contaminants

1. Introduction

Technology continues to play an essential role in solving societal problems across developing countries, including Ghana. This project, titled "IoT Water Quality Monitoring System," aims to develop a practical technological solution that addresses a real-world challenge affecting communities, institutions, and industries. By combining software development techniques with intelligent data processing and monitoring, this system provides users with timely information and decision support. The proposed solution also aligns with global development efforts, such as the United Nations Sustainable Development Goals (SDGs), which emphasize innovation, sustainability, and improved quality of life.

2. Problem Statement

Despite the increasing adoption of digital technologies, many communities still lack access to efficient systems that can monitor environmental risks. Artisanal and small-scale illegal mining, commonly referred to as galamsey, poses a severe threat to local water bodies. Rivers and groundwater sources may appear visually clean yet still contain high concentrations of dissolved heavy metals, cyanides, and acidic runoff. Manual monitoring methods are often prone to human error and may fail to detect early warning signs. Furthermore, many existing solutions are expensive or not designed for the specific needs of developing regions.

3. Gap Analysis: Why This Project is Necessary

To understand the value of this research, it is important to look at the limitations of current methods and organizations. Commercial water monitoring systems, such as those produced by Hach or Xylem, are generally designed for large-scale industrial use. These professional systems are often unavailable or too expensive for widespread use in smaller Ghanaian communities. On the other hand, many existing academic IoT projects, often labeled as Generic Smart Water Systems, focus only on basic parameters like pH and temperature. While these are useful for home aquariums or swimming pools, they are insufficient for detecting the chemical signatures of illegal mining.

Similarly, while remote sensing platforms like SERVIR West Africa can identify where mining is physically taking place through satellite imagery, they cannot provide real-time data on the chemical safety of the water itself. This project bridges these gaps by providing a cost-effective, ground-level detection system that specifically monitors for heavy metals and chemical pollutants through Electrical Conductivity and Oxidation-Reduction Potential. These parameters are essential for identifying mining toxins that basic consumer-grade sensors and satellite monitoring miss.

4. Objectives of the Project

The specific objectives of this project are:

- To study existing technologies and design a functional architecture for a smart monitoring system.
- To develop a working prototype that continuously measures pH, Total Dissolved Solids (TDS), Turbidity, Temperature, Electrical Conductivity (EC), and Oxidation-Reduction Potential (ORP).
- To detect hidden contamination by identifying abnormal patterns in EC and ORP readings, which are key indicators of heavy metal pollution.
- To implement an automated alert system using buzzer notifications and remote messaging for when parameters exceed safe thresholds.
- To design a user-friendly interface that allows for the visualization of real-time and historical data.

5. Methodology and System Architecture

The development follows a structured Software Development Life Cycle (SDLC), moving from requirement analysis to system design and implementation. The system operates on an edge-to-cloud architecture:

- **Data Acquisition:** The ESP32 microcontroller serves as the central processor, reading signals from the multi-sensor array.
- **Onboard Processing:** The system converts raw sensor values into standard units through calibration and noise filtering. A temperature sensor is used to correct other sensor outputs, ensuring accuracy regardless of environmental changes.
- **Transmission:** The ESP32 transmits processed data to a cloud platform via Wi-Fi using communication protocols such as HTTP or MQTT.
- **Cloud and Interface:** The cloud stores the data and applies logic to trigger alerts. A web dashboard then displays the information in charts and graphs for easy monitoring.

6. Tools and Equipment Required

The hardware for this prototype includes the ESP32 Dev Board, which handles data processing and wireless communication. The sensor suite consists of specialized modules for pH, TDS, Turbidity, ORP, and EC, alongside a DS18B20 temperature sensor for calibration. A buzzer module and an OLED display provide local alerts and data viewing. The software stack utilizes C and C++ for the hardware firmware, while the web dashboard is built using HTML, CSS, and JavaScript, with Firebase or ThingSpeak used for real-time data storage.

7. Significance of the Study

The successful implementation of this project will demonstrate how affordable technologies can solve critical problems in developing communities. It provides a system that improves decision-making and operational efficiency for local authorities and residents. By focusing on a simplified and affordable implementation, this project makes high-tech environmental monitoring accessible to the areas that need it most.